Kitwe News Aggregator and Analyzer Project Report

<u>Table of Contents:</u>

Collaborators

Nidha Ahmed Mohammad
Nemat Aloush
Gabriel Blessing
Khadija Chahidi
Chris Coble
Vaishnavi Desai
Mohan Dev Vashisht
Dunni George
Brett Hoy
Fazeel Ahmed Khan
Subash Khanal
Tanishq Kolhatkar
Yulia Khorruzhaya
Jimmy Kimunyi
Joshua Kassim
Ambigapathi Kavin
Yamkela Matolengwe
Edgar Meva
Seifegebreal Mosisa (Seife)
Keith Mwangi
Rosephine Maideyi
Nadia Saoud
Anagha Santhosh
Aditya Sharma
Rohan Singh Negi
Savin Shynu Varghese
Mariela Sanchez
Sorasak Phuangphairoj
Sherilyn Lasrado
Sahishnuta Tosh
Nu Wai Thet
Duncan Otieno
Wamutu Eddie
Colleta Wamalwa

Omdena Zambia, Kitwe Chapter Lead: Lighton Nkandu Kalumba

Abstract

Kitwe, a dynamic city in Zambia, has experienced a significant shift in news consumption patterns with the growing accessibility of the internet. Digital media and social networks now dominate as primary sources of information, enabling real-time access and interaction. However, this digital evolution has also introduced challenges, including the rapid spread of misinformation, which can negatively impact public opinion and community well-being. Local media outlets often struggle to provide timely and accurate coverage amidst the overwhelming flow of information.

To address these challenges, this project proposes the development of a tool powered by a Language Learning Model (LLM) to analyze real-time news data specific to Kitwe. The tool will aggregate information from multiple sources, evaluate credibility, identify trends, and assess public sentiment, presenting insights through a user-friendly interface. By leveraging advanced technology, the initiative aims to enhance the reliability of news dissemination and empower Kitwe's residents to make informed decisions, fostering a more resilient and informed community in the digital age.

Introduction

Globally, fake news has become a significant challenge, impacting societies, politics, economies, and public health. The rapid proliferation of misinformation through social media platforms undermines trust in credible sources, disrupts democratic processes and exacerbates societal divisions. In critical sectors like health, fake news about vaccines and treatments has fueled hesitancy, posing serious public health risks. Detecting and combating fake news is crucial for preserving accurate information flow, maintaining societal harmony, and ensuring informed decision-making.

In Zambia, fake news detection is particularly relevant given the country's growing reliance on social media and digital communication. Locally, misinformation can exacerbate political tensions during elections, disrupt community trust, and incite violence or panic. Public health campaigns, such as those for COVID-19 vaccination, have also faced challenges due to the spread of false information. Additionally, fake news undermines the credibility of media institutions, threatening the role of journalism in holding power to account.

Kitwe, as one of Zambia's largest cities and a hub of economic activity in the Copperbelt Province, holds significant national importance. Given its influence, ensuring the accuracy of news originating from or concerning Kitwe is critical to maintaining public trust, economic stability, and social harmony.

Locally, Kitwe's diverse population relies heavily on social media and digital platforms for information. This high digital penetration creates an environment where fake news can spread rapidly, especially on topics such as political events, labor disputes, or health emergencies. False reports can escalate tensions, such as during labor protests in the mining industry, leading to unnecessary conflicts and economic disruptions.

Verification of Kitwe-specific news is essential to safeguard the city's socio-economic fabric. Reliable information ensures that residents, businesses, and policymakers make informed decisions. Efforts to enhance news verification in Kitwe should include localized fact-checking initiatives, media literacy programs, and partnerships with local news outlets. These steps can help foster a culture of critical thinking and trust in verified information, strengthening Kitwe's resilience against the spread of misinformation.

The news aggregator and analyzer implementation followed a simple, step-by-step process. It started with preparing the data, choosing the right model, training it, and then testing it. This project tackled a major problem: the spread of fake news in Kitwe. False information online is a big issue because it can confuse people, add to the overwhelming amount of online content, and make it hard to know what's true. The goal was to build a news aggregator and analyzer, through a Streamlit application, that could help identify and reduce this problem effectively.

Data Collection

Web Scraping

To collect Kitwe-centered news data, we adopted a systematic approach focused on accessing RSS (Really Simple Syndication) feeds from several Zambian news websites. RSS feeds are standardized XML formats that allow websites to share real-time updates. We applied this approach by configuring a Python script to extract data specifically tagged with “Kitwe” in the RSS feeds, ensuring that only relevant content was gathered. The script iterated over multiple pages from each source, capturing a broad range of articles. This strategy helped us secure a substantial dataset, including content from well-known sources like Lusaka Times, all filtered to prioritize Kitwe.

The data collection process utilized a range of tools and libraries, with Python serving as the primary programming language. The feed parser library was integral to the project, allowing us to extract content from the RSS feeds seamlessly. Additionally, the CSV library in Python was used to store collected data in a structured CSV file, allowing for easy data manipulation and later analysis. The CSV format enabled each article to be saved with detailed attributes, including the news source, category tags, headline, link, description, publication date, and author. This data structure allowed us to maintain organized, comprehensive records for each news entry.

The script was configured to retrieve up to 1,000 pages per news source, maximizing data collection from each outlet. This strategy was instrumental in capturing a wide range of articles, ensuring robust coverage of Kitwe-related news.

Preprocessing and Pipeline Formation

Effective preprocessing is critical for training a large language model (LLM) to function as a fake news detector. This involves robust feature engineering, source credibility analysis, and bias identification to ensure that the model can accurately detect misleading information. Each step of preprocessing contributes to building a more reliable and context-aware system capable of identifying patterns associated with fake news.

Key features extracted from text play a crucial role in enabling the LLM to identify fake news effectively. These features include linguistic cues, sentiment analysis, and source credibility measures. Sentiment analysis, performed using tools like TextBlob, flags articles with extreme emotional tones, such as outrage or excessive happiness, which are often indicators of fake news. Additionally, keyword frequency thresholds are established to detect sensational content, and topic modeling with Latent Dirichlet Allocation (LDA) is used to highlight dubious themes in articles. Headline analysis also plays a part, with clickbait indicators such as excessive capitalization, punctuation, or emotionally charged language flagged as potential red flags.

Source credibility is a foundational element of preprocessing. Using a predefined list of reputable domains and regex patterns, sources are assessed for reliability. Credibility definitions are further contextualized based on regional or cultural nuances, ensuring that sources from both Zambian and international outlets are evaluated appropriately. This process accounts for varying standards of journalism across regions and helps distinguish reputable sources from those more likely to propagate misinformation.

Bias is a critical factor in the spread of fake news, often manifesting in the form of misleading or emotionally charged content. Fake news tends to exploit biases by evoking strong emotional responses and relying on unverifiable sources [1]. Indicators such as clickbait tactics, emotional language, and vague or anonymous authorship are hallmarks of such content [1]. Identifying and analyzing these biases is essential for detecting patterns that contribute to misinformation. Establishing clear guidelines for data analysis helps mitigate these biases, ensuring that unreliable news sources are identified and flagged, ultimately fostering trust within local and global communities.

In summary, specific features designed to analyze credibility are integrated into the preprocessing pipeline. Sentiment analysis detects articles with extreme sentiments, while keyword frequency thresholds flag content that is overly sensational. Topic modeling highlights themes that align with misinformation, and clickbait detection algorithms analyze headlines for patterns indicative of fake news. These credibility-focused features were used in the preprocessing pipeline to classify news as either fake or real.

Methodology

Through usage of preexisting Hugging Face LLM repositories, the data collected and preprocessed was fed to chosen LLMs, trained, and tested. The following models were used in this integration step: ALBERT, BERT, BERT (base-uncased), DeBERTa, DistilBERT, ELECTRA, RoBERTA, and T5. The differences between the models are summarized below:

Model	Description	Key Features/Usage
ALBERT	A model designed to reduce memory consumption and increase training speed.	Parameter sharing and factorized embeddings reduce memory usage; efficient for large datasets.
BERT	The standard pre-trained transformer model for a variety of NLP tasks.	Bidirectional transformer that uses NSP and masked language modeling (MLM); versatile across tasks like classification, QA, and sentiment analysis.
BERT (bert base-uncased)	A pre-trained language model by Google that uses bidirectional training to understand context by analyzing words in both directions. With 12 transformer layers and 110 million parameters, it excels in natural language processing tasks like text	BERT-Base Uncased is widely used for applications such as text classification, sentiment analysis, question answering, and named entity recognition, and is known for setting benchmarks in NLP tasks.

	classification, sentiment analysis, and question answering	
DeBERTa	A model with disentangled attention mechanisms for better handling of complex dependencies in text.	Disentangled attention separates content and positional information; improved contextual understanding for text.
DistilBERT	A lighter version of BERT optimized for speed and efficiency.	Reduced size and faster inference compared to BERT, suitable for resource-constrained environments.
ELECTRA	A language model developed by Google that employs a discriminator approach (similar to GANs) to differentiate between real and fake tokens [2]	Efficient pre-training by replacing masked language modeling with token replacement tasks [2]; fine-tuned for text classification using Hugging Face uncased model.
RoBERTa	An optimized version of BERT with improved training techniques.	Removes Next Sentence Prediction (NSP) and trains on more data with larger batches for improved performance.
T5	A pre-trained Text-to-Text Transfer	Versatile for text generation,

	Transformer (T5) that follows an encoder-decoder architecture suited for sequence-to-sequence tasks [3].	summarization, and classification; smaller variant with 60M parameters used for resource efficiency.

After comparative testing, BERT(bert-base-uncased) was chosen for its balance between contextual accuracy and computational efficiency leading to the best performance in the model evaluation. The evaluation metrics and individual model's performances are listed in the evaluation and results section.

To address class imbalance during training, Focal Loss was used. This made the model place greater weight on harder-to-classify examples, especially from the minority class. Early stopping was applied to halt training as soon as validation loss stopped improving, preventing overfitting. Quantization techniques were also applied to reduce the model's size subtly by converting parameter weights from floats to integers. This decrease in memory usage consequently makes the app easier to deploy.

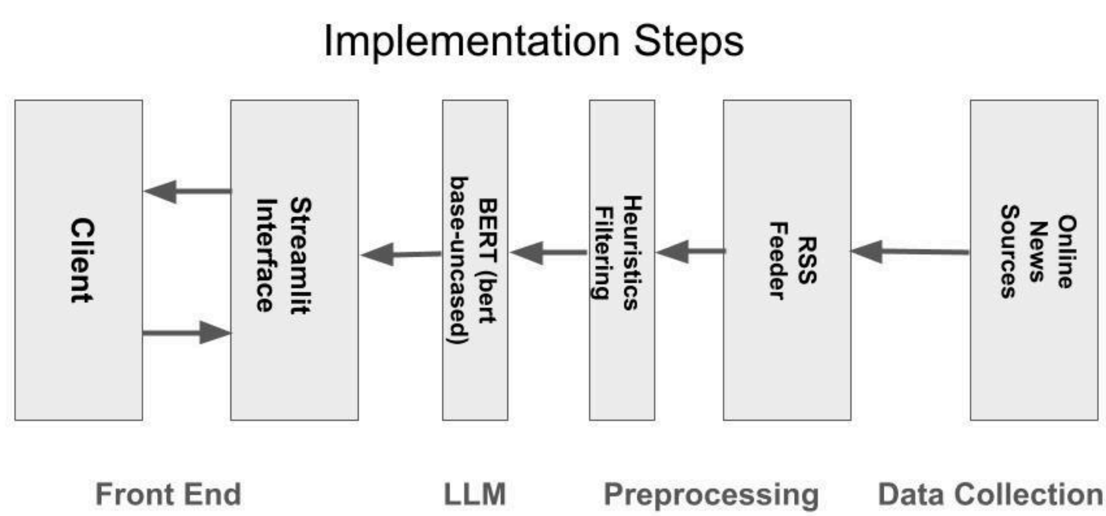
The training was conducted on Google Colab utilizing a T4 GPU, which provided the necessary computational power for handling the transformer-based architecture. The GPU was opted over the CPU because it allows the LLM to train faster.

Implementation

Deploying the trained LLM to the Streamlit application ensured seamless integration and optimal functionality for the user. The model was first uploaded to a Hugging Face repository, allowing for streamlined access and management within the Streamlit framework. The LLM was then loaded into the application script to enable efficient processing. The Streamlit interface was designed with two primary features: a news aggregator and a news analyzer. The aggregator provides a live news feed sourced from the data used during training, with each article automatically classified as either real or fake. The analyzer offers a user-driven option to input articles for evaluation, allowing users to submit a title and content to receive a credibility classification, while adding the inputs as a new data point. This application holds significant potential for aiding the people of Kitwe by promoting informed decision-making and helping identify credible news sources in real time.

The development team refined the application through multiple updates to optimize the user experience. The news aggregator page features the thirty most recent articles curated by the system. Each article includes a clear visual indicator of its authenticity, marked by color coding—red for fake and green for real. Users can further organize their feed by sorting articles based on category or news source. To enhance the application's branding and convey a sense of commitment to its purpose, the team designed a distinctive logo, depicting a scene of Kitwe's influential copper mining.

Below is a schematic of the steps leading to the model's implementation:



Evaluation and Results

Below is a table summarizing the different LLMs used in the integration step, their key features that describe their operation, and their respective performances in terms of accuracy, precision, recall, and F1 score:

Model	**Accuracy**	**Precision**	**Recall**	**F1 Score**
ALBERT	0.79	0.80	0.73	0.76
BERT	0.80	0.80	0.75	0.78
BERT (bert base-uncased)	**0.98**	**1.00**	**0.98**	**0.99**
DeBERTa	0.78	0.74	0.82	0.78
DistilBERT	0.80	0.79	0.78	0.78
ELECTRA	0.88	0.79	0.88	0.84
RoBERTa	0.79	0.74	0.81	0.78
T5	0.82	0.82	0.82	0.82

BERT (bert base-uncased) was chosen due to having the best performance in accuracy, precision, recall, and F1 score. The hyperparameter tuning used for this model was a learning rate of 2e-5, a batch size of 16, two epochs, and the data is split into 80% for training and 20% for testing.

Discussion

Data Collection Discussion

During the data collection phase, several challenges arose. One key challenge was inconsistent data availability. Some RSS feeds did not consistently provide data for every field, leading to gaps in critical information like publication dates or descriptions. Additionally, pagination proved complex; some sources used irregular pagination, causing the script to prematurely stop retrieving data in certain cases.

Tagging and categorization presented further issues. Many articles lacked detailed tags or categories, complicating our attempts to classify news by topic. Specifically, generalizing an approach to categorizing news articles was very subjective in many cases making it challenging to agree upon. Connection issues also emerged periodically, with interruptions in accessing RSS feeds causing delays. Although retry mechanisms were implemented to handle these interruptions, not all sources provided stable connections.

Duplicate entries were another notable issue. Since RSS feeds occasionally repeated articles across pages, we needed to filter for duplicates, a step crucial to maintaining data integrity. These challenges highlighted areas for potential improvement in both the script and the data collection process.

There were limitations that impacted the comprehensiveness of the dataset. One limitation was our reliance on RSS feeds; the data collection was restricted to articles that were available through these feeds. If a news item was not indexed by the RSS feed, it would not appear in our dataset. This reliance may have led to incomplete data coverage.

Another limitation was the occurrence of incomplete entries. Some articles lacked full descriptions or sufficient metadata, making deeper analysis more challenging. Additionally, since the scope was centered around Kitwe, there is an inherent source bias in the data. This project targeted Kitwe news specifically, meaning that broader regional news coverage was not included.

Finally, while automated processes handled most of the data retrieval, some manual quality checks were necessary. These checks, though essential, introduced the potential for human error.

Exploratory Data Analysis Discussion:

The preprocessed dataset reveals a significant imbalance between real and fake news articles, with 8128 articles classified as real news and 4218 as fake news. This disparity poses challenges for training machine learning models, as imbalances can lead to biased predictions. To address this, techniques such as resampling or adjusting class weights during training are essential to ensure that the model treats both classes fairly and avoids favoring the overrepresented real news category.

One of the key trends in the dataset is the dominance of politics as a category for both real and fake news. This overrepresentation could introduce biases into classification models, as they might become overly attuned to political content at the expense of other categories. On the other hand, the narcotics category is significantly underrepresented, which may limit the model's ability to generalize effectively for this type of content.

For real news, articles are most frequently categorized under politics, career, and business, reflecting traditional media trends. In contrast, fake news exhibits a more even distribution across categories, although it still shows a heavy concentration in politics. These patterns underline the importance of careful feature engineering and model tuning to avoid skewed predictions based on category dominance.

Analyzing temporal trends reveals fascinating insights. Real news articles show a sharp decline after 1985, which may point to shifts in reporting practices or changes in media consumption. Meanwhile, fake news articles surged dramatically from the late 2000s to 2020, potentially reflecting the rise of misinformation campaigns and shifts in how fake news is produced and disseminated.

The pre-2020 spike in fake news is particularly striking and suggests deliberate efforts to spread misinformation during critical time frames. This underscores the importance of incorporating temporal features into the model to capture these trends effectively.

Patterns in publication times offer additional insights into the behavior of real and fake news. Real news articles are more prevalent on weekdays, likely aligning with traditional news publishing cycles. Fake news, however, tends to spike over the weekend, particularly from Friday to Sunday, which may indicate targeted misinformation strategies designed to exploit audiences that are less critical or more relaxed during these days.

Hourly patterns further highlight these differences. Fake news peaks around the 10th hour of the day, suggesting coordinated efforts to release misinformation at a specific time. Real news, by contrast, exhibits a more even distribution throughout the day. Additionally, Thursdays see a noticeable dip in fake news activity, which could correspond to specific reporting or misinformation trends.

Looking at the broader trends, real news shows a consistent decline over the years, possibly reflecting changing dynamics in journalism or audience preferences. Fake news, in contrast, is more volatile, responding reactively to current events and demonstrating its opportunistic nature. These contrasting patterns highlight the need for models to adapt dynamically to such fluctuations to maintain accurate predictions across varying contexts.

Methodology Discussion:

In reiteration of the exploratory data analysis performed, the dataset reveals an imbalance between real (8128) and fake (4218) news articles, which could lead to biased machine learning predictions. Techniques like resampling or adjusting class weights are necessary to mitigate this issue. Politics dominates both real and fake news categories, potentially biasing models, while the narcotics category is underrepresented, limiting model generalization. Real news is predominantly about politics, career, and business, while fake news is more evenly distributed but still focused on politics. Temporal trends show a decline in real news after 1985 and a surge in fake news from the late 2000s to 2020, reflecting misinformation campaigns. Fake news peaks on weekends and around 10 AM, with noticeable drops on Thursdays. These patterns emphasize the need for careful feature engineering and model adaptation to account for category dominance, temporal trends, and publication behavior.

The decision to use Hugging Face models like BERT, RoBERTa, and ALBERT is because they are already highly effective for text classification tasks, such as fake news detection, due to their strong contextual understanding. These models, pre-trained on vast datasets, require minimal fine-tuning for specific tasks, saving training time. Their "uncased" versions simplify tokenization by ignoring case sensitivity, reducing complexity. Additionally, they benefit from transfer learning, utilizing pre-trained weights to reduce the need for large labeled datasets and enabling more efficient model training and tuning.

Implementation Discussion:

After training, the large language model (LLM) was uploaded to the Hugging Face Hub, which served as a centralized model registry and deployment platform. This step ensured that the model was easily accessible and version-controlled for seamless integration into other applications. The Hugging Face API allowed the Streamlit app to access the model through the transformers library, specifically leveraging the pipeline function to streamline user interaction. This integration provided a straightforward interface where users could input news articles and receive predictions on whether the content was real or fake.

To optimize inference speed and improve the app's efficiency, a hybrid approach was employed to preprocess and filter content before sending it to the LLM. Incoming news items were first analyzed using heuristic-based filters designed to flag suspect content. These heuristics considered factors such as excessive use of buzzwords, abnormally short text sequences, or unusually high word counts, which are common characteristics of fake news. Only the articles that triggered these flags were sent to the model for further evaluation, significantly reducing computational overhead and improving the responsiveness of the application.

Additionally, caching mechanisms were implemented to enhance the app's performance further. Both predictions and previously scraped news articles were stored in a cache to prevent redundant processing during repeated queries. This ensured that users received results almost instantaneously for articles that had already been evaluated. Combined with the streamlined integration of the model, these optimizations created an efficient, user-friendly fake news detection tool that balanced accuracy and performance.

Future Work:

Future work on this project will focus on several enhancements to improve its effectiveness and expand its capabilities. First, increasing the number of data sources will help address the current imbalance in the dataset, such as the disparity between fake (30%) and real (60%) news articles, ensuring the model is better trained for balanced classification. Additionally, bias identification will be extended by incorporating more indicators, allowing for a deeper understanding of underlying trends and improving the system's capacity to flag biased content. Updating the model architecture and fine-tuning processes will further minimize misclassification, enhancing the overall accuracy and reliability of the predictions.

Moreover, integrating the Streamlit application with other web tools like Flask APIs will open new possibilities for feature expansion. Flask can handle the backend logic, enabling more robust functionalities while maintaining a user-friendly interface. This integration will support dynamic interaction with external databases, seamless API usage, and additional analytical tools, making the application more versatile and scalable for real-world use.

Conclusion

In conclusion, addressing the spread of fake news is paramount, especially in a context like Kitwe, where digital reliance amplifies the risks of misinformation. The successful implementation of a news aggregator and analyzer using a fine-tuned BERT model demonstrates a promising step toward combating this issue. Despite challenges such as imbalanced datasets, incomplete entries, and inconsistent data from RSS feeds, the project highlighted key trends, including the prevalence of politically-oriented misinformation and temporal patterns in fake news dissemination. By leveraging advanced machine learning techniques and incorporating measures such as resampling and feature engineering, the system effectively balanced class predictions and captured critical contextual nuances, reinforcing its utility in identifying and mitigating misinformation.

Moving forward, expanding the dataset to include broader regional news and enhancing data collection methods will address some limitations and improve the comprehensiveness of analysis. Coupling this with localized fact-checking initiatives, media literacy programs, and collaboration with news outlets will help sustain the impact of this tool. By integrating these strategies, Kitwe can strengthen its resilience against the socio-economic disruptions caused by misinformation and foster a culture of informed decision-making, ensuring the city's stability and progress in an increasingly digital age.

References

Literature:

1. Bias in news guidelines: https://utopia.ut.edu/FakeNews/factcheck
2. Electra article: https://openreview.net/pdf?id=r1xMH1BtvB
3. T5 article: https://arxiv.org/abs/1910.10683
4. Quantization introduction: https://medium.com/@techresearchspace/what-is-quantization-in-llm-01ba61968a51
5. General overview of using BERT in NLP: https://arxiv.org/abs/2103.11943
6. Paper detailing the usage of regularization in preexisting LLMs: https://arxiv.org/abs/1911.03437
7. Thesis paper on using dropout to improve neural networks: https://www.cs.toronto.edu/~nitish/msc_thesis.pdf
8. Paper on optimizing training loss in hyperparameter tuning in NLP: https://aclanthology.org/D15-1253.pdf

News Sources:

- Daily Mail Zambia
- Flava FM
- Christian Voice
- Copperbelt Energy
- ZNBC
- News Invasion 24
- Lusaka Time
- Kitwe Online
- Daily Revelation Zambia
- Zambia Monitor
- Tech Africa News
- Daily Nation Zambia
- Lusaka Star
- Lusaka Voice
- Mwebantu
- Zambia365
- Zambia Eye
- Zambia Report

Toolstack:

1. **NLTK, NumPy, and Pandas**: For preprocessing operations such as tokenization, cleaning, and feature extraction.
2. **Datasets library**: To store and manage training data effectively.
3. **Pytorch:** To construct LLMs.
4. **Matplotlib and Seaborn**: Visualization for EDA.
5. **Hugging Face Transformers**: For model fine-tuning and repository access.
6. **Scikit-learn**: To compute evaluation metrics.
7. **Streamlit:** Built an interactive and user-friendly web application for deploying the model and performing real-time predictions.

Jupyter Notebooks:

1. Data Collection and Preprocessing Pipeline Notebooks: https://github.com/EdBali/KitweNewsProject
2. Fazeel Ahmed Khan EDA Notebook: Colab Notebook
3. Subash Khanal EDA Notebook: Dagshub Notebook
4. Tanishq Kolhatkar EDA Notebook: Colab Notebook
5. Sorasak Phuangphairoj EDA Notebook: Colab Notebook
6. Joshua Kassim EDA Notebook: Colab Notebook
7. Nadia Saoud EDA Notebook: Dagshub Preprocessing Pipeline
8. Rosephine Maideyi EDA Notebook: Dagshub Notebook
9. Sahishnuta Tosh EDA Notebook: GitHub Repository
10. Chris Coble EDA Notebook: Dagshub Notebook
11. Anagha Santhosh EDA Notebook: Dagshub Repository
12. Mohan Dev Vashisht EDA Notebook: Dagshub Notebook
13. LLM Implementation Notebook from Subash Khanal (Initial): https://colab.research.google.com/#fileId=https%3A//dagshub.com/kalumbalighton/KitweChapter_LocalNewsAggregatorAnalyzer/raw/d7c760e21562ebc97acb26714e04f3ddb796629d/LLM_bert_quantized_input.ipynb
14. LLM Implementation Notebook from Mariela Sanchez: https://dagshub.com/kalumbalighton/KitweChapter_LocalNewsAggregatorAnalyzer/src/main/Test_LLM_Electra_KitweNews.ipynb
15. LLM Implementation Notebook from Subash Khanal (Final): https://colab.research.google.com/drive/1l5LK5qCN-e4FUj4fdoFgGkE0j1mY8evf?usp=sharing
16. LLM Implementation Notebook : https://dagshub.com/kalumbalighton/KitweChapter_LocalNewsAggregatorAnalyzer/src/main/t5_small_fine_tuning.ipynb

Hugging Face Repositories:

Model	Hugging Face link
DistilBert	Distilbirt repo
ALBERT	ALBERT Repo
RoBERTa	RoBERTa Repo
DeBERTa	DeBERTa Repo
BERT	BERT repo
ELECTRA	ELECTRA repo
T5	T5 repo
BERT (bert base-uncased)	Base Uncased Repo

Streamlit Application Link:

https://kitwenewstoday.streamlit.app/